

Viaggio.ai

AI-Powered Unified Travel Planning Platform

Product Requirements & Technical Architecture Document

Executive Summary

Viaggio.ai is a conversational AI travel planning platform designed to solve the fragmentation problem in travel booking. Rather than forcing users to navigate multiple websites for flights, hotels, tours, trains, and car rentals, Viaggio.ai consolidates the entire travel planning experience into a single, intelligent interface.

Core Innovation: Combines natural language AI interaction with comprehensive travel inventory aggregation, with special emphasis on tours, activities, and train travel—components typically overlooked by mainstream booking platforms.

Problem Statement

Primary Pain Point: Travel planning requires juggling multiple platforms (Expedia for flights, Booking.com for hotels, Viator for tours, various sites for trains), leading to:

- Fragmented user experience and decision fatigue
- Difficulty coordinating schedules across different bookings
- Missed opportunities for complete itinerary planning
- Particularly problematic for international travelers needing trains and local experiences

Target Audience

Primary Market: North American travelers (United States & Canada)

Rationale:

- Highest travel expenditure for international trips

- Longest distances to reach Europe, Asia, and other international destinations
- Higher average travel budgets
- English as primary language simplifies initial development
- Strong market for comprehensive travel planning tools

Core Features

Travel Components

Phase 1 (MVP):

1. Flights
2. Hotels
3. Tours & Activities

Phase 1.5 (Quick Follow-up):

4. Train Tickets (high priority for international travel)

Phase 2:

5. Car Rentals

Conversational AI Capabilities

Users interact with an enthusiastic travel expert AI agent that can:

- Understand natural language travel requests
- Search and aggregate results from multiple providers
- Answer refinement questions:
 - "Are there any airports closer?"
 - "Show me only [specific airlines/hotels/providers]"
 - "Can you find cheaper options?"
 - "I need a window seat"
 - "Make it more family-friendly"
- Compile final itineraries from user selections
- Remember preferences across the session

Itinerary Experience

When users request "Put together my itinerary," the system generates a comprehensive travel plan:

Display Format

- Modal/popup window in current session
- Day-by-day breakdown of all travel components
- Flight details (dates, times, airlines, confirmation numbers)
- Hotel bookings (check-in/out, location, amenities)
- Tours scheduled chronologically
- Train connections with departure/arrival times
- Total cost breakdown by category
- Optional: embedded map visualization

Available Actions

- **Add to Cart:** Saves selections for potential booking
- **Export to PDF:** Generates downloadable formatted itinerary
- **Share:** (Future) Email or link sharing capability

Technical Architecture

Frontend

- React-based conversational interface
- Claude API integration for AI agent
- Session storage for temporary data persistence
- Itinerary builder with PDF export functionality
- Mobile-responsive design
- Modern/minimal UI with professional, welcoming aesthetic

Backend (Future Development Required)

- Node.js with Express for API orchestration
- Redis caching system to minimize API costs
- Rate limiting for expense control
- Affiliate tracking integration
- PostgreSQL database (Phase 2 - user accounts)

Hosting

- Vercel for frontend hosting
- Railway or Render for backend services
- Estimated cost: \$50-100/month

API Integration Strategy

Component	Provider	Cost Model	Notes
Flights	Amadeus	\$0.35-0.50/search	~500-700 searches/month with budget
Hotels	Booking.com	FREE (Affiliate)	4-6% commission per booking
Tours	Viator	FREE (Affiliate)	8% commission, 300k+ tours
Tours	GetYourGuide	FREE (Affiliate)	8-10% commission
Trains	Trainline / Rail Europe	Partnership required	5-10% commission, B2B partnerships
Car Rentals	Rentalcars.com	FREE (Affiliate)	50% revenue share on margin

Budget Allocation (\$500/month)

- Amadeus (Flights): \$175-250
- Claude API: \$50-100
- Hosting/Infrastructure: \$50-100
- Reserve/Buffer: \$100

Revenue Model

Commission Structure

- Hotels: 4-6% per booking
- Tours: 8-10% per booking
- Car Rentals: 50% margin share
- Trains: 5-10% (when partnerships established)
- Flights: 1-2% (affiliate links to airlines/OTAs)

Example Economics

Assumptions:

- 100 users book monthly
- Average booking value: \$1,500
- Average commission: 5%

Monthly Revenue: ~\$7,500

Profit (after \$500 API costs): ~\$7,000

Development Roadmap

Phase 1: Prototype

Timeline: Immediate (1 day)

Features:

- Conversational UI with AI agent
- Simulated search results
- Itinerary builder with PDF export
- Responsive design

Purpose: Test UX, validate concept, demo to investors

Phase 2: Backend Development

Timeline: 2-3 months

Cost: \$5,000-15,000 (freelancer) or DIY

Features:

- Real API integrations (flights, hotels, tours)
- Caching layer for cost optimization
- Affiliate tracking system
- Rate limiting

Phase 3: Beta Launch

Timeline: +1 month testing

Features:

- Limited user base
- Monitor API costs and conversion rates
- Gather user feedback
- Iterate on UX

Phase 4: Scale & User Accounts

Timeline: +2-3 months

Features:

- User authentication system
- Saved preferences and trip history
- Email notifications
- Cross-session memory

Key Differentiators

- **Conversational AI:** Natural language interaction vs. form-based searches
- **Comprehensive Coverage:** Tours and train tickets alongside traditional components
- **Iterative Refinement:** Ongoing dialogue to perfect trip details

- **Unified Itinerary:** Single view of entire trip across all providers
- **International Focus:** Optimized for North American travelers going abroad

Risks & Mitigation Strategies

API Access Challenges

Risk: Train APIs require B2B partnerships that may be difficult to secure initially

Mitigation: Start with affiliate links to train booking sites, pursue partnerships as platform grows

Cost Management

Risk: API costs could escalate quickly with user testing

Mitigation: Implement aggressive caching, rate limiting per user, monitor usage closely

Market Competition

Risk: Crowded market with established players (Kayak, Skyscanner, Hopper)

Mitigation: Focus on AI conversational experience and comprehensive itinerary building as key differentiators

Legal Compliance

Risk: API terms of service, affiliate agreements, data privacy regulations

Mitigation: Review all provider terms carefully, consult legal counsel before launch, implement GDPR-compliant data practices

Sample Use Case

User Request:

"Party of two want to go to Florence, Italy in September for 7 days. Don't want to spend more than \$2,000 on flights and \$1,000 on hotel."

AI Agent Response Process:

1. Confirms details and preferences (departure city, specific dates, airline preferences)
2. Searches flight options under \$2,000 for two passengers
3. Presents top 3-5 flight options with different airlines/connections
4. User selects preferred flight
5. Searches hotels in Florence under \$1,000 for 7 nights
6. Presents hotels by neighborhood with amenities
7. User asks: "Show me options in the historic center with breakfast included"
8. AI refines results
9. User selects hotel
10. AI suggests: "Would you like me to find tours and activities in Florence?"
11. Presents Uffizi Gallery tickets, cooking classes, day trips to Tuscany
12. User selects 2-3 activities
13. User requests: "Put together my final itinerary"
14. System generates day-by-day itinerary with all bookings, times, locations
15. User exports to PDF for offline reference

Brand Identity: Viaggio.ai

Name Origin

"Viaggio" means "journey" or "travel" in Italian, evoking European travel and adventure

Design Philosophy

- **Modern/Minimal:** Clean interfaces, plenty of white space, clear typography
- **Professional:** Conveys trust and reliability of an established company
- **Welcoming:** Warm colors that psychologically draw users in
- **Easy Navigation:** Intuitive flow, minimal clicks to accomplish tasks

AI Agent Personality

Enthusiastic Travel Expert: Knowledgeable, excited about travel, helpful without being pushy, uses conversational language while maintaining professionalism

Immediate Next Steps

1. Build interactive prototype with simulated data
2. Test itinerary generation with and without map integration
3. Gather initial user feedback on conversational flow
4. Apply for affiliate partnerships (Booking.com, Viator, GetYourGuide)
5. Begin Amadeus API developer account setup
6. Refine UI/UX based on prototype testing
7. Develop backend architecture specifications
8. Source developer resources for Phase 2 implementation

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